Information Theory, Coding and Inference — 2026

Exam 3: April 2026

Instructor: Shashank Vatedka

Instructions: *This is a closed-book exam. You are not permitted to refer to any material or discuss the problem with anyone. Malpractice will be severely punished. Please mention your ROLL Number and name clearly in the answer sheet.*

Justify all your statements clearly. You may use any result proved in class (but clearly state which results you are using), but everything else needs to be proved. Total time: 80 minutes

Question 3.1. The mutual information

$$I(X;Y) = I(p_X, p_{Y|X})$$

is ______ (convex/concave/neither) function of $p_{Y|X}$ for a given p_X, and a ______ (convex/concave/neither) function of p_X for a given $p_{Y|X}$ (1pt+1pt) — no justification required.

Prove that the rate-distortion function $R^*(D)$ is nonincreasing and convex in D. (2pts+8pts) — you may use the above without proof (if required).

Question 3.2. Let $p_{Y_1|X_1}$ be a channel with input alphabet $\mathcal{X}_1$ and output alphabet $\mathcal{Y}_1$, and $p_{Y_2|X_2}$ be a channel with input alphabet $\mathcal{X}_2$ and output alphabet $\mathcal{Y}_2$. Let C_1 and C_2 be the capacities of the two channels. We want to say something about the capacity of the overall channel obtained by combining the channels in different ways.

For each of the following, specify the input alphabet (in terms of $\mathcal{X}_1, \mathcal{X}_2$), output alphabet (in terms of $\mathcal{Y}_1, \mathcal{Y}_2$), and the capacity (in terms of C_1, C_2).

1. For each channel use/time instant, the transmitter is allowed to choose any one of the two channels to transmit in. For eg, it may transmit symbols $x(1), x(2) \in \mathcal{X}_1$ in the first two channel uses (i.e., first two time instants), $x(3) \in \mathcal{X}_2$ in the third channel use, $x(4) \in \mathcal{X}_1$ in the fourth, and so on. At any time instant, if it transmits a symbol $x(t)$ in channel i (for $i \in \{1,2\}$) then the receiver gets a symbol $Y(t) \in \mathcal{Y}_i$ distributed according to $p_{Y_i|X_i}(\cdot|x(t))$. (1pt+1pt+5pts)
2. In each channel use/time instant, the transmitter is allowed to send two symbols (x, x'): one across channel 1 and another across channel 2. The receiver receives two symbols (Y, Y'), where $Y \sim p_{Y_1|X_1}(\cdot|x)$ and $Y' \sim p_{Y_2|X_2}(\cdot|x')$, and these are conditionally independent given (x, x'). (1pt+1pt+8pts)
3. Cascade channel: Now, assume that $\mathcal{X}_2 = \mathcal{Y}_1$, and the output of channel 1 is fed to the input of channel 2. For this case, show that the overall capacity is at most $\min\{C_1, C_2\}$. (1pt+1pt+3pts)

Question 3.3. Derive the capacity of the BSC(p). (6pts)

Consider a cascade channel where the first channel is BSC(p) whose output is fed to the input of the second channel which is BSC(q). Find the capacity of this cascade channel. (4pts)